

ORGANIZING THE CONTENT

Information Architecture and Application structure patterns.

Navigation, Signposts and wayfinding patterns

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Post Requirements Gathering

- At this point you should know what your users want out of the application.
- What's Next?
 - General Mental Pictures, Sketches, Mock ups?
- High Level Organization is often the most difficult task in designing a user interface
- So before you get into the design, separate the content from it's physical presentation
 - Don't start by thinking of things in terms of Windows, Tabs, Panels, Controls, etc.
 - Start by organizing actions and object abstractly

Order of Design Operations

1) Organization and Task Flows

- Separate Actions and Objects from Presentation

2) Physical Structure

- Layout Windows, Panels, Controls, etc

- Difference in general web page design vs. application design?

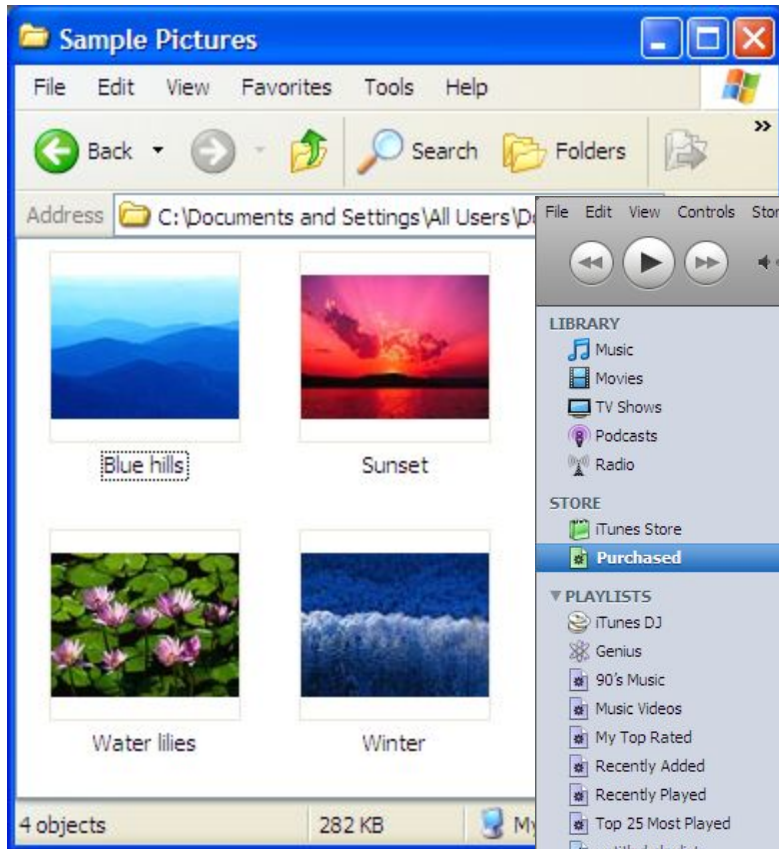
Common Application Organization Approaches

- Lists of Objects
 - Inbox, Task List, Shopping Cart, etc
- Lists of Actions or Tasks
 - Browse, Buy, Sell, Write, Draw, etc
- List of Subject Categories
 - Genre, Type, etc
- List of Tools
 - Calendar, Address Book, Notepad, etc

Considerations for Choosing and Approach

- The nature and domain of the application
- Users' domain knowledge
- Users' comfort level with technology
- Comparable mental models pre-existing for users in your domain
- Your decision here can make or break an interface
- Is it ok to mix paradigms?

Lists of Objects



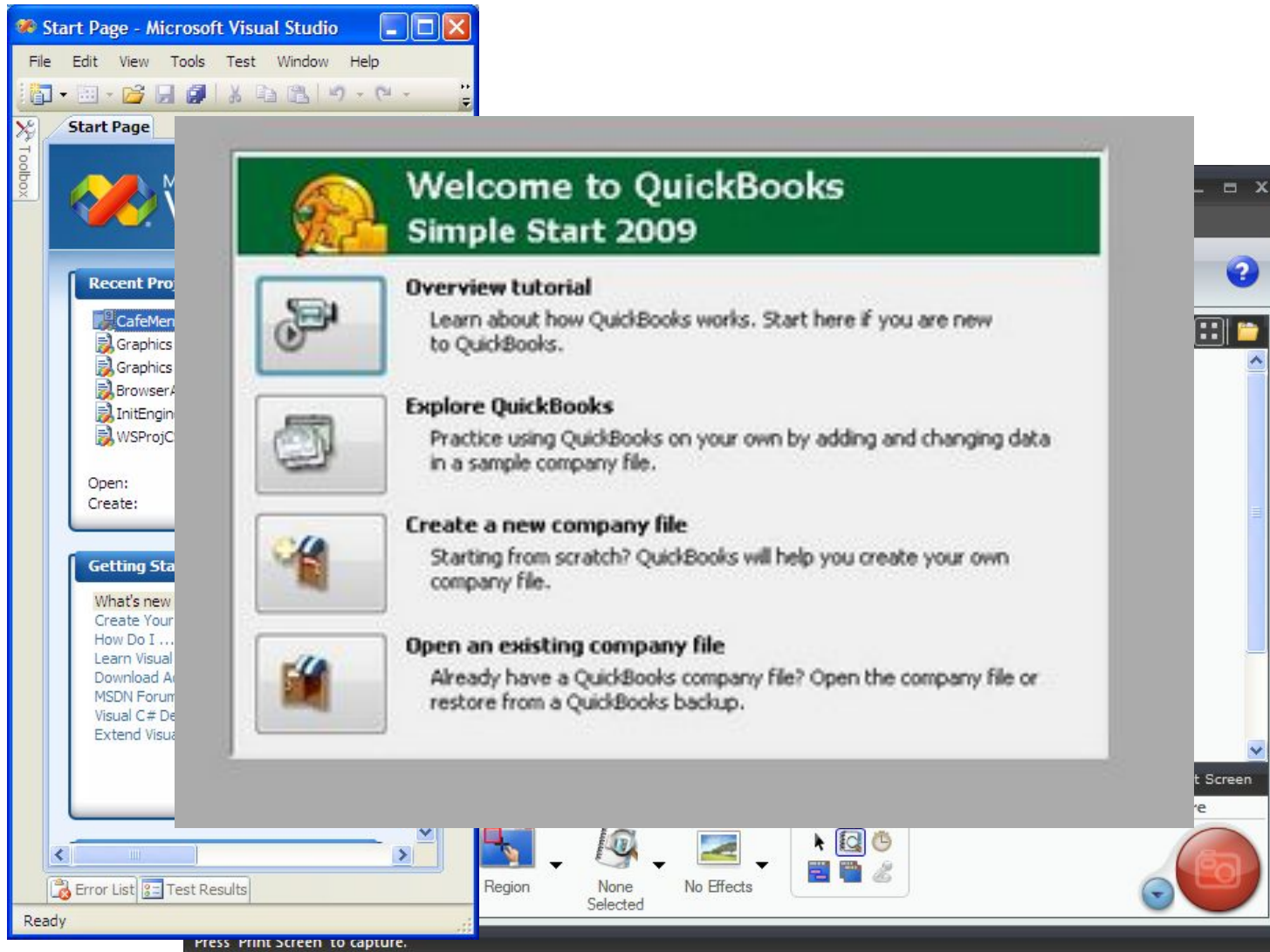
List of Objects

- Often the most obvious user interface approach
- Examples
 - Email, Songs, Books, Images
 - Search Results, Financial Transactions
 - Contacts, Tasks, Explorers
- From these we reach familiar UI Idioms:
 - Forms to Edit or View things
 - Players or Viewers to play media
 - Web Pages to Print things

Presenting Lists of Objects

- Common Models
 - Linear (typically sortable)
 - Tables (typically sortable by column and filterable)
 - Hierarchies
 - Groups items by Category
 - Groups items by relationship (parent/child)
 - Spatial (Maps, Charts, etc)
- Controls
 - Tables, Lists, Trees, etc

List of Actions



List of Actions

- Verb-centered rather than noun-centered
- “What do you want to do?”
- “What do you want to work on?”
- “What type of item would you like to create?”
- Benefit is often that this type of interface is explained in plain English (plain language)
- How many actions are appropriate?

Presenting Lists of Actions

- Single Window Drilldown
- Engage Selected Item's Action
- Double Click to Invoke
- Other Approaches (not high level):
 - Buttons, Menu Bars, Pop-up Menus
 - Dropdown Menus, Toolbars, Links, Action Panels
- Real-estate or efficiency concerns?

List of Categories

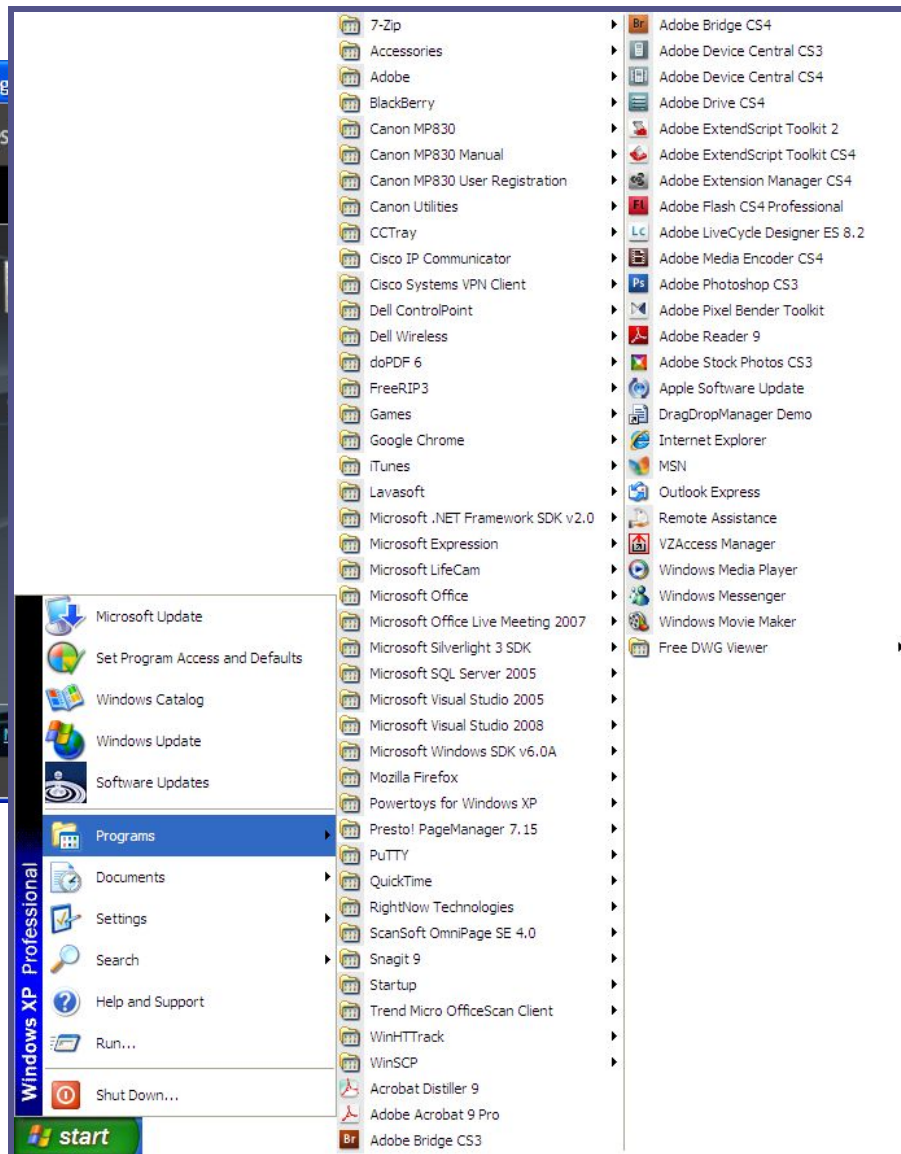
The image shows a screenshot of a web browser window displaying the ESPN website. The browser's address bar shows the URL <http://www.espn.go.com/>. The ESPN website features a red header with the ESPN logo and navigation links for TV, Radio, and Magazine. Below the header, there are sections for "ALL SPORTS", "COLUMNISTS", "PAGE 2", and "FANTASY". The main content area includes "TOP STORIES" and "TOP VIDEOS: Featuring Iverson Has A". A large image of a Pittsburgh Steelers player is visible, with the headline "Getting Offensive" and the subtext "Defense dominated ... until the first half's final two minutes. Steelers and Titans tied 7-7. GameCast LIVE! Blog".

Overlaid on the right side of the browser window is the iTunes application window. The iTunes window shows the "LIBRARY" tab selected in the sidebar. The "PLAYLISTS" section is highlighted with a red box, and the "90's Music" playlist is selected. The main area of the iTunes window displays a list of songs with columns for Name, Time, Artist, and Album. The list includes songs such as "All 4 Love" by Color Me Badd, "Blaze of Glory" by Jon Bon Jovi, "How's It Going to Be" by Third Eye Blind, "Life Is a Highway" by Tom Cochrane, "Cherry Pie" by Warrant, "You've Lost That Lovin' Feelin'" by Righteous Brothers, "Black Hole Sun" by Soundgarden, "High Enough" by Damn Yankees, "Secret Garden" by Bruce Springsteen, "Let's Hear It for the Boy" by Deniece Williams, "Faith" by George Michael, "Freedom" by George Michael, "Footloose" by Kenny Loggins, "Danger Zone" by Kenny Loggins, "Sweet Caroline" by Neil Diamond, "Life Is a Highway" by Tom Cochrane, "Cherry Pie" by Warrant, "Bohemian Rhapsody" by Queen, "Another One Bites the Dust" by Queen, "Baby Got Back" by Sir Mix A Lot, "Born to Be Wild" by Steppenwolf, "Build Me Up Buttercup" by The Foundations, "Celebration" by Kool & The Gang, "Cotton Eye Joe" by Rednex, and "Dancing Queen" by ABBA.

	Name	Time	Artist	Album
21	☒ All 4 Love	3:32	Color Me Badd	C.M.B.
22	☒ Blaze of Glory	5:35	Jon Bon Jovi	Blaze of Glory (In
23	☒ How's It Going to Be	4:12	Third Eye Blind	Third Eye Blind: A
24	☒ Life Is a Highway	4:27	Tom Cochrane	Mad Mad World
25	☒ Cherry Pie	3:21	Warrant	Cherry Pie
26	☒ You've Lost That Lovin' Feelin'	3:45	Righteous Brothers	The Very Best of
27	☒ Black Hole Sun	5:18	Soundgarden	Superunknown
28	☒ High Enough	4:45	Damn Yankees	Damn Yankees
29	☒ Secret Garden	4:27	Bruce Springsteen	Bruce Springsteen
30	☒ Let's Hear It for the Boy	4:22	Deniece Williams	Footloose (15th
31	☒ Faith	3:13	George Michael	Ladies & Gentlem
32	☒ Freedom	6:31	George Michael	Listen Without Pr
33	☒ Footloose	3:47	Kenny Loggins	Footloose (15th
34	☒ Danger Zone	3:35	Kenny Loggins	Yesterday, Toda
35	☒ Sweet Caroline	3:22	Neil Diamond	The Neil Diamond
36	☒ Life Is a Highway	4:27	Tom Cochrane	Mad Mad World
37	☒ Cherry Pie	3:21	Warrant	Cherry Pie
38	☒ Bohemian Rhapsody	5:55	Queen	Classic Queen
39	☒ Another One Bites the Dust	3:36	Queen	Queen: Greatest
40	☒ Baby Got Back	4:23	Sir Mix A Lot	Mack Daddy
41	☒ Born to Be Wild	3:31	Steppenwolf	Steppenwolf: All
42	☒ Build Me Up Buttercup	2:55	The Foundations	Back to the 60's,
43	☒ Celebration	4:59	Kool & The Gang	Funk Essentials:
44	☒ Cotton Eye Joe	3:14	Rednex	Sex & Violins
45	☒ Dancing Queen	3:52	ABBA	Gold - Greatest

Lists of Categories

- Used often in website design due to the large amount of diverse data mean for many people
- Not used often in application design. Categories are better for noun oriented software rather than action oriented applications.
- That doesn't mean categories aren't a good tool for organizing subsets of an applications presentation data



Lists of Tools

- Commonly used by Operating Systems, Palm Pilots, Blackberrys, Cell Phones, etc
- Some Applications use this paradigm as well: Microsoft Money, Quick Books, etc
- Tool-based Organization typically fails when the names of tools mix with actions, tasks or objects – User expectations

Presenting Lists of Tools

- Linear is most common
 - Flat list
 - Table
- Sorted?
 - Alphabetically
 - By Category
 - By Frequency of Use
 - Customizable
- *Hub and Spoke Navigation*

Navigation, Signposts and Wayfinding

A series of horizontal lines of varying lengths and shades of gray and white, extending from the left edge of the slide and branching out towards the right, creating a modern, architectural feel.

Staying Found

- Good Signage
- Environmental Clues
- Map

Page Depth

- Keep navigation paths short (number of pages)
- Avoid nested dialogs/windows
- Functionality vs. Usability Tradeoffs

Patterns

- Clear Entry Points
- Global Navigation
- Hub and Spoke
- Pyramid
- Modal Panel
- Sequence Map
- Breadcrumbs
- Annotated Scrollbar
- Color-Coded Sections
- Animated Transition
- Escape Hatch

Navigation Patterns

A series of horizontal lines of varying lengths and shades of gray and white, extending from the right edge of the slide and overlapping the white area below the title.

Clear Entry Points

- Present a few task oriented entry points into an interface

The screenshot displays the ING DIRECT website interface. At the top left is the ING DIRECT logo with the tagline "Save your money". To the right are links for "SIGN IN", "OPEN ACCOUNT", and "HELP". On the left side, a dark blue vertical bar contains three white text links: "View my account", "Open an account", and "Learn more", each accompanied by an orange sphere icon. The main content area is titled "Save Your Money" in large blue font. Below this title, there are three distinct savings products, each featuring an orange sphere icon and a blue arrow pointing to a description:

- Easy OrangeSM**: Low rate 3.99% (3.98% APR). Description: "Welcome home, savers".
- shareBUILDER[®]**: Invest with any amount. Description: "Yep. Even \$100".
- Orange CDSM**: Elevate your savings. Description: "1.85% APY 12 month CD".

The footer is orange and contains the following information:

- Savers' Blog with icons for YouTube, Facebook, and Twitter.
- Visit our Cafés: New York | Philadelphia | Los Angeles | Wilmington, DE | Chicago | St. Cloud, MN | Honolulu.
- MEMBER FDIC logo.
- EQUAL HOUSING LENDER logo.

Clear Entry Points?

E*TRADE®

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GO

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TRADING TOOLS

RESEARCH & GUIDANCE

RETIREMENT

BANKING

PRICING

OPEN AN ACCOUNT

GENERATE INCOME & MANAGE RISK

**DO MORE
WITH OPTIONS**

We'll show you how.

First Step: Upgrade Your Account in Minutes

GET STARTED



E*TRADE SECURITIES LLC

SECURE LOG ON

User ID: Password:

Start In:

Accounts

LOG ON

中文

Digital Security ID token expired?

Forgot your User ID or Password

Set up Online Account Access

Get Quotes Enter Symbol(s) or Name

GO

Markets Overview | News | Charts

DOW 9771.91
-48.29 (-0.49%)

NASDAQ 2132.91
+0.05 (+0.00%)

S&P 500 1062.73
-5.57 (-0.52%)

10 Yr. T-Note 3.446%
-0.28

1:07 PM ET 9/21/09 15 min delay

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2009 SmartMoney

Global Navigation

- Reserve a section of every page to show a consistent set of links or buttons to navigate to key sections of the application (or site)



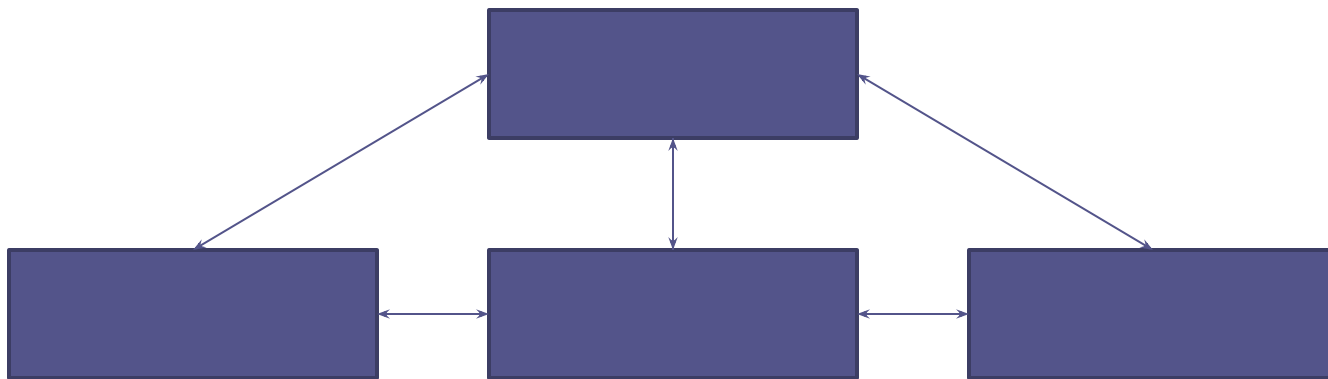
Hub and Spoke

- Sub Applications reached from the main navigation page with one way in and one way out.



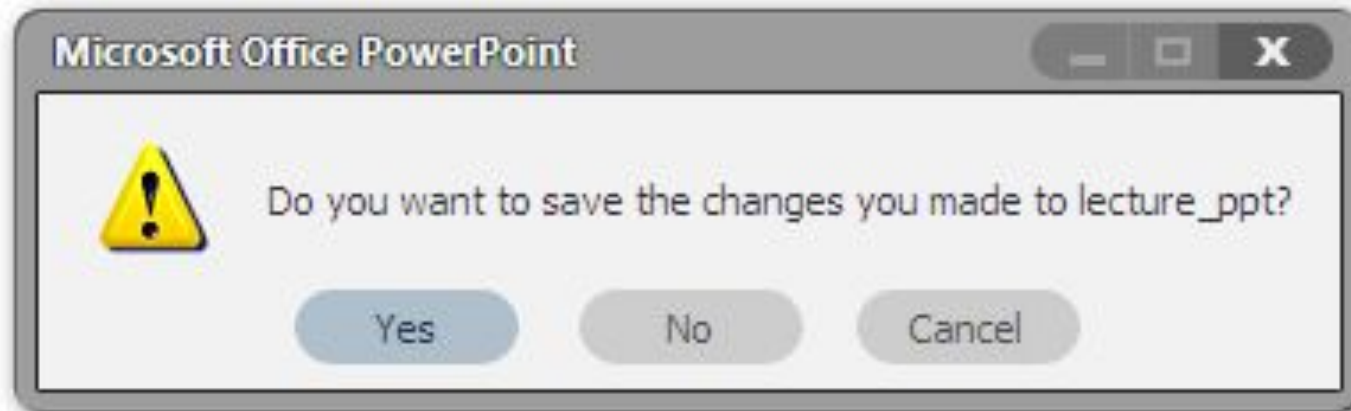
Pyramid

- Linked Sequence of Pages with Previous/Back and Next links or Actions
- Typically combined with a main page that has navigation links to each page individually



Modal Panel

- Show one page with navigation options only to accomplish the immediate task



Signposts

A series of horizontal lines of varying lengths and shades of gray and white, extending from the left edge of the slide and overlapping the white area below the title.

Sequence Map

- Show the full navigation map on each page with indication of current position in the sequence
- Navigation: Map markers are typically clickable as a method to jump forward or backward in the sequence

Breadcrumbs

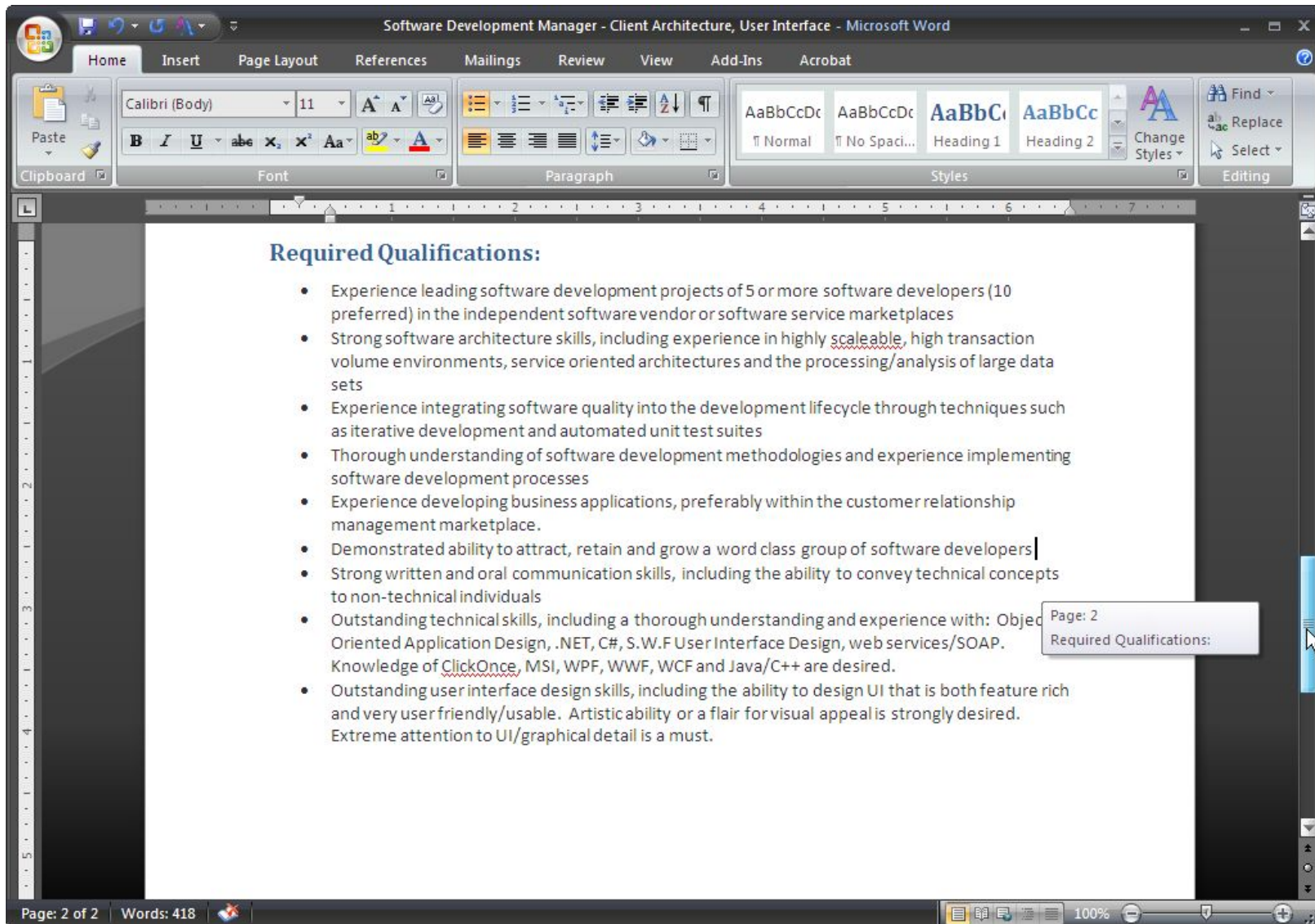
- Show a map of current and all parent/previous pages on current page
- Navigation: Parent/Previous pages are typically clickable navigation links to return up the chain



Annotated Scrollbar

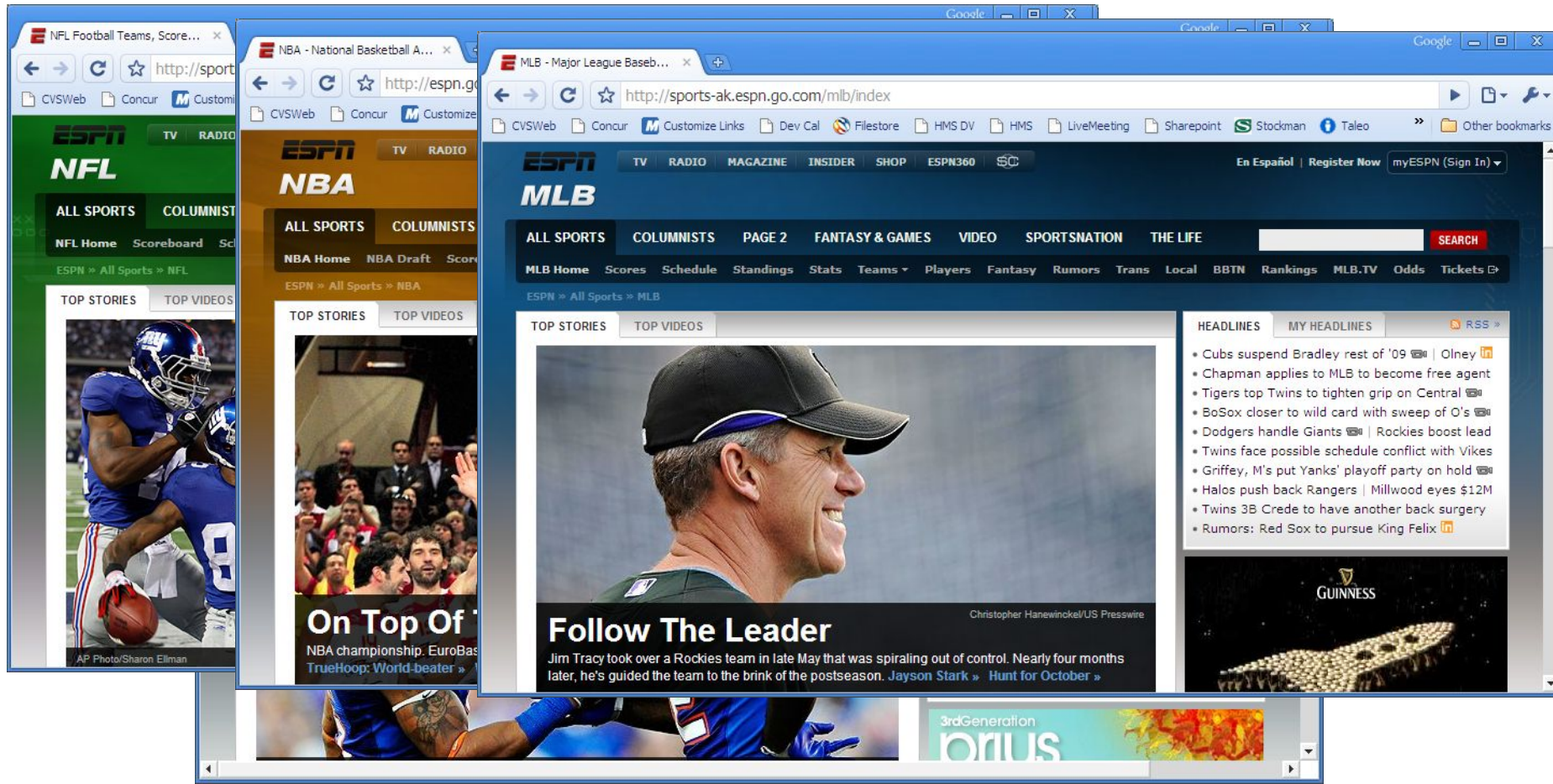
- Use the scrollbar to act as a current location notification in your sequence map
- Navigation: Scrollbar itself

Annotated Scrollbar



Color Coded Sections

- Use of color to identify sequence or location of current page in an application



Misc

A series of horizontal lines of varying lengths and shades of gray and white, extending from the left edge of the slide towards the right, positioned below the 'Misc' text.

Animated Transition

- Use of animation to assist a user in understanding page transition
- Typically used when navigation path is a large leap or complex path for the user to comprehend

Escape Hatch

- Pages with limited navigation options/actions should have an action that aborts the current operation and returns to a known place

